

Victor Gusev

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Education

University of California, Los Angeles (UCLA)

Expected Graduation: June 2028

B.S. in Computer Science and Engineering

GPA: 3.64 / 4.00

Coursework: Algorithms, Object-Oriented Design, Programming Languages, Probability, Logic Design

Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript, Assembly

Robotics & AI: ROS, NVIDIA Isaac Lab, PyTorch, PID Control, LiDAR, Computer Vision, Sensor Integration

Frameworks & Web: React, Node.js, Express, Socket.io, MongoDB

Software, Tools & Hardware: Git, Docker, Linux, Solidworks, 3D Printing

Experience

Software Engineering Intern | [Dimaag.ai](https://dimaag.ai)

June 2026 – August 2026

- Architected a hybrid Reinforcement Learning and classical local planner for autonomous vehicles to improve accuracy and speed of travel along complex curved paths
- Implemented PPO algorithm training pipelines, 2D LiDAR sensor raycasting, and domain randomization within NVIDIA Isaac Lab
- Bridged the sim-to-real gap by validating policies across various simulators, implementing domain randomization to ensure seamless hardware deployment.
- Validated and deployed autonomous navigation policies on physical hardware, achieving reliable tracking at >12 mph with sub-decimeter trajectory accuracy
- Integrated advanced LLM tooling into the engineering pipeline to automate architectural documentation, generate robust system test cases, and parse complex training logs

Robotics Programming Lead | [FIRST Robotics Program](#)

August 2021 – May 2025

- Led software development for a 15–20 member robotics team, building semi-autonomous competition robots
- Implemented PID control systems, vision tracking (Limelight), and LiDAR-based localization
- Collaborated with mechanical and electrical leads to safely deploy and test software on the physical robot hardware
- Engineered autonomous routines and vision pipelines that propelled the team to the FIRST World Championships, ranking in the top 5% of regional competitors.

Projects

Proof | React, MongoDB, Web Scraping, Gemini API

- Architected a full-stack progressive web application (PWA) with MongoDB and React to version-control recipe iterations and manage real-time inventory states
- Integrated the Gemini API and web scraping (Cheerio) to autonomously extract, intelligently restructure, and import recipes directly from raw URLs
- Engineered a real-time, cross-device timer synchronization system using Socket.io to track concurrent baking stages across multiple platforms
- Maintained application stability by creating over 100 automated test cases to prevent bugs during new feature deployments.
- Link: <https://proof-cdvj.onrender.com>

Micromouse Simulator | Java, Graphics, File Parsing

- Designed a full maze-solving simulator for Micromouse competitions using Java
- Implemented Flood Fill pathfinding algorithm to autonomously solve unknown mazes
- Built a graphical visualization engine using Java Graphics to animate traversal and decisions
- Developed file parsing pipeline to load and replay real competition mazes
- GitHub: github.com/victor2275/micromousesim